MSTAR PRACTICE WORKSHEET

Lesson 6-12: Least Common Multiple

Grade 6 Mathematics · Standard 6.NOS.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

What is the LCM of 3 and 5?

A 3

B 5

C 15

D 30

Multiples of 3: 3, 6, 9, 12, 15, ... Multiples of 5: 5, 10, 15, ... The smallest common multiple is 15.

Question 2 SELECTED RESPONSE — CORRECT: C

What is the LCM of 6 and 8?

A 6

B 14

C 24

D 48

Multiples of 6: 6, 12, 18, 24, ... Multiples of 8: 8, 16, 24, ... The smallest common multiple is 24.

Question 3 SELECTED RESPONSE — CORRECT: C

What is the LCM of 4 and 10?

- A 2
- B 10
- C 20
- D 40

> Multiples of 4: 4, 8, 12, 16, 20, ... Multiples of 10: 10, 20, ... The smallest common multiple is 20.

Question 4 SELECTED RESPONSE — CORRECT: D

What is the LCM of 2 and 7?

- A 2
- B 7
- C 9
- D 14

> Since 2 and 7 share no common factors (they are relatively prime), LCM = $2 \times 7 = 14$.

Question 5 SELECTED RESPONSE — CORRECT: C

Two buses leave the station at 8:00 AM. Bus A returns every 15 minutes and Bus B returns every 20 minutes. When is the next time both buses are at the station together?

- A 8:30 AM (30 minutes later)
- B 8:40 AM (40 minutes later)
- C 9:00 AM (60 minutes later)
- D 9:20 AM (80 minutes later)

> LCM(15, 20) = 60 minutes. Both buses return together 60 minutes after 8:00 AM, at 9:00 AM.

Question 6 SELECTED RESPONSE — CORRECT: C

One machine runs every 4 min, another every 6 min. They just ran together. In how many minutes do they run together again?

- A 6
- B 10
- C 12
- D 24

> Skip count: multiples of 4 are 4, 8, 12; multiples of 6 are 6, 12. The first shared multiple is LCM(4, 6) = 12 minutes.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

What is the least common multiple of 6 and 8?

- (A) 2
- (B) 14
- (C) 24
- (D) 48

> Multiples of 8 are 8, 16, 24, 32, and multiples of 6 are 6, 12, 18, 24. The first multiple that appears in both lists is 24.

- **A:** That is the greatest common FACTOR. A common multiple is larger than both numbers, not smaller.
- **B:** That adds 6 and 8. A common multiple has to divide evenly by both.
- **D:** That is 6 times 8, a common multiple but not the LEAST one. 24 appears in both lists first.

Part B — correct answer: D

Which reasoning shows why your answer to Part A is correct?

- (A) Multiply the two numbers together, $6 \times 8 = 48$, because the product is always the least common multiple.
- (B) Find the largest number that divides both 6 and 8, which is 2.
- (C) Add the two numbers, $6 + 8 = 14$.
- (D) Skip count starting with the larger number: 8, 16, 24. Then skip count by 6: 6, 12, 18, 24. The first non-zero multiple in both lists is 24.

> The least common multiple is the first shared multiple in the two skip-counting lists. The product 48 is a common multiple but not the least one, and 2 is the greatest common factor.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Multiplying Instead of Listing

Sofia found the least common multiple of 4 and 10 by multiplying the two numbers: $4 \times 10 = 40$. She wrote that the least common multiple of 4 and 10 is 40.

Explain Sofia's misconception and give the correct least common multiple.

Model answer: The product 40 is a common multiple, but it is only the least one when the two numbers share no factor other than 1. Here 4 and 10 both have the factor 2. Skip count from the larger number: 10, 20. Skip count by 4: 4, 8, 12, 16, 20. The first multiple in both lists is 20, so the least common multiple is 20.

2 points	Student explains that the product is a common multiple but not always the least one because 4 and 10 share the factor 2, and gives 20 as the least common multiple.
1 point	Student gives 20 as the least common multiple but does not explain why the product is too large.
0 points	Student agrees with Sofia that the least common multiple is 40 or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

You are making party favors. Glow sticks come in packs of 10 and bracelets come in packs of 4. What is the fewest of each you can buy so every favor has exactly one glow stick and one bracelet with none left over? Explain how the LCM solves this problem.

Model answer: Every favor uses one glow stick and one bracelet, so the totals must match and be a multiple of both 10 and 4. The LCM of 10 and 4 is 20, so you buy 2 packs of glow sticks ($2 \times 10 = 20$) and 5 packs of bracelets ($5 \times 4 = 20$) and finish with 20 complete favors and nothing left over.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.